8 Puzzle Problem Using Python

Aim

To solve the 8 Puzzle Problem using Python.

Objective

To arrange the tiles in the correct order by moving the blank space until the goal state is reached.

Algorithm

1. Define the initial puzzle state.
2. Define the goal state.
3. Find the position of the blank tile (0).
4. Move the blank tile in possible directions (Up, Down, Left, Right).
5. Generate the next possible states.
6. Repeat until the goal state is reached.
7. Display the solution.

Flowchart

```
    Start
      |
      v
Input Initial State
      |
      v
Compare with Goal State
   /       \
 Yes        No
  |          |
```

```
Print Solution   Generate
     |          Possible Moves
     |               |
     +------<--------+
          |
          v
        Stop
```

Code

```python
                print()
            break

        visited.add(state)

        for i in range(3):
            for j in range(3):
                if state[i][j] == 0:
                    x, y = i, j

        for dx, dy in moves:
            nx, ny = x + dx, y + dy

            if 0 <= nx < 3 and 0 <= ny < 3:
                new = [list(row) for row in state]
                new[x][y], new[nx][ny] = new[nx][ny], new[x][y]
                new_state = tuple(tuple(row) for row in new)

                if new_state not in visited:
                    queue.append((new_state, path + [state]))
```

Output

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

Goal Reached!
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

(1, 2, 3)
(4, 5, 6)
(7, 8, 0)
```

Result

The 8 Puzzle Problem was solved successfully, and the goal state was reached using Python.

Conclusion

The program demonstrates how the 8 Puzzle Problem can be solved by generating valid moves until the goal state is achieved. This illustrates the basic concept of state-space search in Artificial Intelligence.